

# 20251102 Only Anticorrelation in DFD13

In short, I believe under the assumptions and conditions made by [DFD13](#), there only exists anti-correlation between SFR and  $Z$ . The anti-correlation from DFD13 is independent on the SFH. However, [WL21](#) loosen some restrictions and produce both correlation and anti-correlation, so next week we should focus on the connection between DFD13 and WL21.

## General bathtub model

Here we start with two continuity equations:

$$\begin{aligned}\frac{dM_g[t]}{dt} &= \Phi[t] - (1 - R)\text{SFR}[t] - \lambda\text{SFR}[t] \\ &= \Phi[t] - (1 - R + \lambda)\text{SFR}[t]\end{aligned}\quad (1)$$

Here  $M_g[t]$  is the gas mass,  $\Phi[t]$  is the inflow rate,  $R$  is the returned fraction from stars and  $\lambda$  is the mass loading factor. So these 3 terms are gas mass changes due to inflow, loss to star formation and loss to outflow.

$$\frac{dM_Z[t]}{dt} = y(1 - R)\text{SFR}[t] - Z[t](1 - R + \lambda)\text{SFR}[t] + Z_{\text{in}}\Phi[t] \quad (2)$$

Here  $M_Z[t] = Z[t] \cdot M_g[t]$  is the metal mass,  $y$  is the metal yield per stellar generation and  $Z_{\text{in}}$  is the metallicity of the inflow gas. Again, these 3 terms are metal mass changes due to gain from metal production, loss to star formation and gain from inflow.

Now we can expand Eq. 2 by product rule to get

$$Z[t]\frac{dM_g[t]}{dt} + M_g[t]\frac{dZ[t]}{dt} = y(1 - R)\text{SFR}[t] - Z[t](1 - R + \lambda)\text{SFR}[t] + Z_{\text{in}}\Phi[t] \quad (3)$$

And substitute the Eq. 1 to equation above and simplify it to get  $\frac{dZ[t]}{dt}$

$$\begin{aligned}Z[t](\Phi[t] - (1 - R + \lambda)\text{SFR}[t]) + M_g[t]\frac{dZ[t]}{dt} &= y(1 - R)\text{SFR}[t] - Z[t](1 - R + \lambda)\text{SFR}[t] + Z_{\text{in}}\Phi[t] \\ M_g[t]\frac{dZ[t]}{dt} &= y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z[t])\Phi[t] \\ \frac{dZ[t]}{dt} &= \frac{1}{M_g[t]}(y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z[t])\Phi[t])\end{aligned}$$

Therefore, we end up with the following equations for the general bathtub model:

$$\begin{cases} \frac{dM_*}{dt} = \text{SFR}[t] = \epsilon[t]M_g[t] \\ \frac{dM_g}{dt} = \Phi[t] - (1 - R + \lambda)\text{SFR}[t] \\ \frac{dZ[t]}{dt} = \frac{1}{M_g[t]}(y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z[t])\Phi[t]) \end{cases} \quad (5)$$

## DFD13

In DFD13, they make some the assumptions:

1. The inflow rate  $\Phi[t]$  is proportional to  $\text{SFR}[t]$  by a constant  $a$ :

$$\Phi[t] = a\text{SFR}[t] \quad (6)$$

2. The star formation efficiency ( $\epsilon$ ), return fraction ( $R$ ), mass loading factor ( $w$ , note that they use  $w$  rather than  $\lambda$ ) and metal yield per stellar ( $y$ ) are also constant.
3. The inflow metallicity is set to be 0:

$$Z_{\text{in}} = 0 \quad (7)$$

Thus, the equations of DFD13 becomes

$$\begin{cases} \frac{dM_*}{dt} = \text{SFR}[t] = \epsilon M_g \\ \frac{dM_g}{dt} = (R - 1 + a - w)\text{SFR}[t] \\ \frac{dZ[t]}{dt} = y(1 - R)\epsilon - aZ[t]\epsilon \end{cases} \quad (8)$$

First, we check the analytical solution of  $\text{SFR}[t]$ :

$$\begin{aligned} \frac{dM_g}{dt} &= (R - 1 + a - w)\text{SFR}[t] \\ \Leftrightarrow \frac{1}{\epsilon} \frac{d\text{SFR}[t]}{dt} &= (R - 1 + a - w)\text{SFR}[t] \\ \Leftrightarrow \frac{d\text{SFR}[t]}{\text{SFR}[t]} &= \epsilon(R - 1 + a - w)dt \\ \Leftrightarrow \text{SFR}[t] &= \text{SFR}_0 e^{\epsilon(R-1+a-w)t} \\ &= \text{SFR}_0 e^{-kt} \end{aligned} \quad (9)$$

Here we define

$$k = \epsilon(1 - R + w - a) \quad (10)$$

and  $\text{SFR}_0$  is the initial SFR at  $t = 0$ .

Then, we check analytical solution of  $Z[t]$ :

$$\begin{aligned} \frac{dZ[t]}{dt} &= y(1 - R)\epsilon - aZ[t]\epsilon \\ \Leftrightarrow Z[t] &= \frac{y(1 - R)}{a} + (Z_0 - \frac{y(1 - R)}{a})e^{-a\epsilon t} \end{aligned} \quad (11)$$

Therefore, under the assumptions from DFD13, SFR and  $Z$  have to satisfy the following forms

$$\begin{cases} \text{SFR}[t] = \text{SFR}_0 e^{-kt} \\ Z[t] = \frac{y(1 - R)}{a} + (Z_0 - \frac{y(1 - R)}{a})e^{-a\epsilon t} \end{cases} \quad (12)$$

Clearly, we have exponential accumulation of  $Z$ .

Since  $k$  is normally negative, this leads to exponential decay of SFH.

Under a special condition that

$$k = 0 \Leftrightarrow a = 1 - R + w \quad (13)$$

this leads to constant SFR.

Also, we further assume the initial metallicity is 0, we have

$$\begin{cases} \text{SFR}[t] = \text{SFR}_0 e^{-kt} \\ Z[t] = \frac{y(1 - R)}{a}(1 - e^{-a\epsilon t}) \end{cases} \quad (14)$$

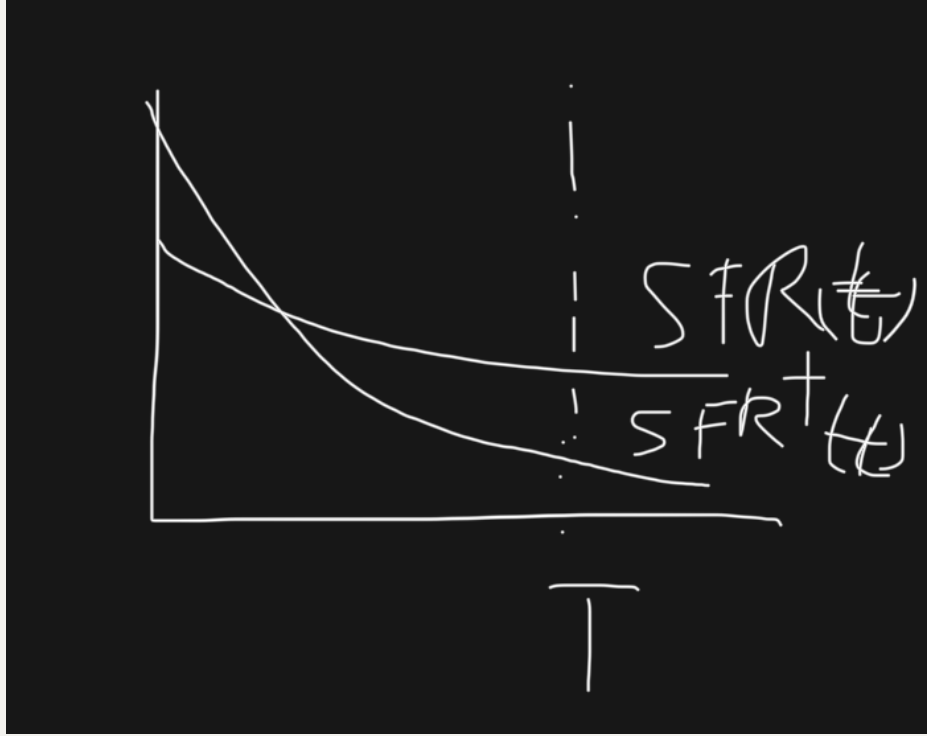
Now with these two equations, we can construct different SFHs to explore the relations between SFR and  $Z$ . The idea is that, given two different  $\text{SFR}_1[t]$  and  $\text{SFR}_2[t]$ , under the condition that at moment  $T$  they build up same stellar mass

$$\int_0^T \text{SFR}_1[t] dt = M_* = \int_0^T \text{SFR}_2[t] dt \quad (15)$$

then we want to check  $\text{SFR}_1[T]$  &  $\text{SFR}_2[T]$  and  $Z_1[T]$  &  $Z_2[T]$ .

Below I describe 5 different scenarios, which I believe should include all the possible cases of SFH.

### DFD13 Case 1



Consider the case that we have two exponential decay SFH. Compared to the default  $\text{SFR}[t] = \text{SFR}_0 e^{-kt}$ , we define a more expotential decay SFH that

$$\text{SFR}^\dagger[t] = \alpha \text{SFR}_0 e^{-\beta kt} \quad (16)$$

with  $1 < \alpha < \beta$ , so that they satisfy

$$\begin{aligned} M_* &= \int_0^T \text{SFR}[t] dt = \int_0^T \text{SFR}^\dagger[t] dt \\ \Leftrightarrow \int_0^T \text{SFR}_0 e^{-kt} dt &= \int_0^T \alpha \text{SFR}_0 e^{-\beta kt} dt \\ \Leftrightarrow \frac{\text{SFR}_0}{k} (1 - e^{-kT}) &= \frac{\alpha \text{SFR}_0}{\beta k} (1 - e^{-kT}) \\ \Leftrightarrow 1 - e^{-kT} &= \frac{\alpha}{\beta} (1 - e^{-kT}) \end{aligned} \quad (17)$$

Obviously, we have  $\text{SFR}[T] > \text{SFR}^\dagger[T]$ .

For metallicity, we can express  $a$  with  $k$  for default case:

$$\begin{aligned}
Z[t] &= \frac{y(1-R)}{a} (1 - e^{-a\epsilon t}) \\
&= \frac{y(1-R)}{1-R+w-\frac{k}{\epsilon}} (1 - e^{-(1-R+w-\frac{k}{\epsilon})\epsilon t})
\end{aligned} \tag{18}$$

Then for the more exponential decay case, we can simply replace  $k$  with  $\beta k$ :

$$\begin{aligned}
Z^\dagger[t] &= \frac{y(1-R)}{1-R+w-\frac{\beta k}{\epsilon}} (1 - e^{-(1-R+w-\frac{\beta k}{\epsilon})\epsilon t}) \\
&= \frac{y(1-R)}{1-R+w-\beta(1-R+w-a)} (1 - e^{-(1-R+w-\beta(1-R+w-a))\epsilon t}) \\
&= \frac{y(1-R)}{a\beta - (\beta-1)(1-R+w)} (1 - e^{-(a\beta - (\beta-1)(1-R+w))\epsilon t})
\end{aligned} \tag{19}$$

Now we define

$$\begin{cases} x = \epsilon t > 0 \\ z = a\beta - (\beta-1)(1-R+w) \end{cases} \tag{20}$$

Thus, we can simplify both metallicity to be

$$\begin{cases} Z(T) = \frac{1 - e^{-ax}}{a} \\ Z^\dagger(T) = \frac{1 - e^{-zx}}{z} \end{cases} \tag{21}$$

Since we have

$$\begin{aligned}
(\beta-1)(1-R+w-a) &> 0 \\
\Leftrightarrow (\beta-1)(1-R+w) &> a(\beta-1) \\
&\Leftrightarrow a > a\beta - (\beta-1)(1-R+w) \\
&\Leftrightarrow a > z
\end{aligned} \tag{22}$$

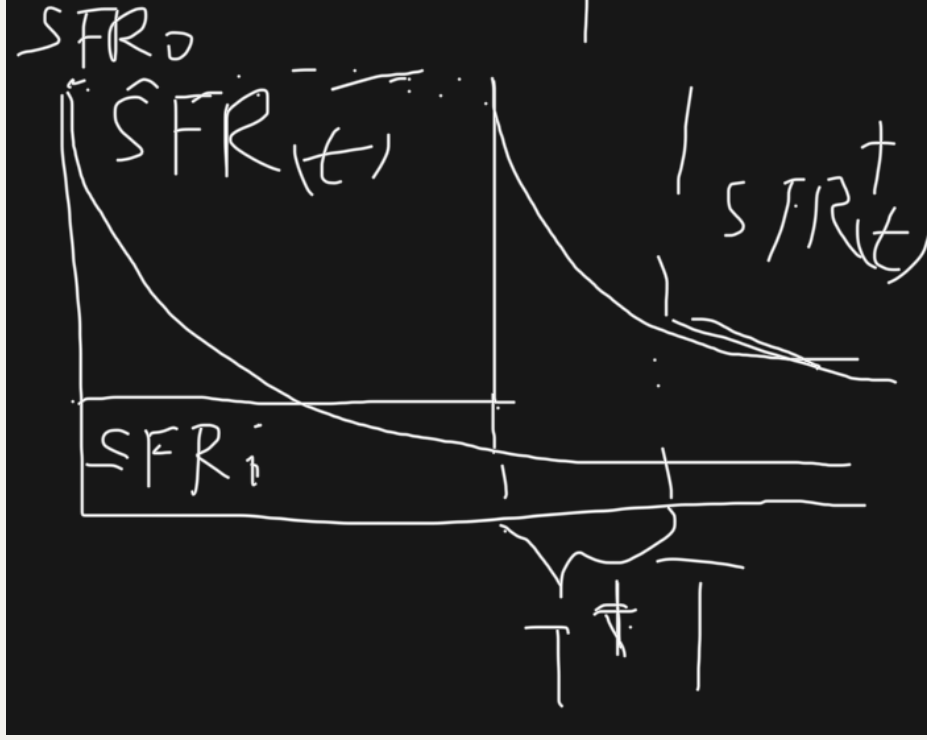
Then we also get

$$Z^\dagger(T) = \frac{1 - e^{-zx}}{z} > \frac{1 - e^{-ax}}{a} = Z(T) \tag{23}$$

Therefore, in case 1, we have the anti-correlation:

$$\begin{cases} \text{SFR}[T] > \text{SFR}^\dagger[T] \\ Z^\dagger(T) > Z(T) \end{cases} \tag{24}$$

DFD13 Case 2



In the case, we keep the default  $\text{SFR}[t] = \text{SFR}_0 e^{-kt}$ , and then we create a SFH that experiences a constant  $\text{SFR}_i < \text{SFR}_0$  first and then a delay exponential decay:

$$\text{SFR}^\dagger[t] = \begin{cases} \text{SFR}_i, & t < T - T^\dagger \\ \text{SFR}_0 e^{-kt}, & t \geq T - T^\dagger \end{cases} \quad (25)$$

Note that

$$\begin{cases} \text{SFR}_i = \epsilon_i M_{g,0} \\ \text{SFR}_0 = \epsilon_0 M_{g,0} \end{cases} \quad (26)$$

where  $M_{g,0}$  is the initial gas mass.

Suppose that at the time  $t = T$ , they both reach the same  $M_*$ :

$$\begin{aligned} M_* &= \int_0^T \text{SFR}[t] dt = \int_0^T \text{SFR}^\dagger[t] dt \\ &\Leftrightarrow \int_0^T \text{SFR}_0 e^{-kt} dt = \text{SFR}_i (T - T^\dagger) + \int_0^{T^\dagger} \text{SFR}_0 e^{-kt} dt \\ &\Leftrightarrow \frac{\text{SFR}_0}{k} (1 - e^{-kT}) = \text{SFR}_i (T - T^\dagger) + \frac{\text{SFR}_0}{k} (1 - e^{-kT^\dagger}) \\ &\Leftrightarrow e^{-kT^\dagger} - e^{-kT} = \frac{k \text{SFR}_i (T - T^\dagger)}{\text{SFR}_0} \\ &= \frac{k \epsilon_i (T - T^\dagger)}{\epsilon_0} \end{aligned} \quad (27)$$

Obviously, we have  $\text{SFR}[T] < \text{SFR}^\dagger[T]$ .

Then we look at the metallicity in the delay case. During the period that  $t \in [0, T - T^\dagger]$ , a constant SFR implies

$$\begin{aligned} \frac{dM_g}{dt} &= 0 \\ &\Leftrightarrow a = 1 - R + w \end{aligned} \quad (28)$$

Hence, the metallicity of the constant SFH will be

$$Z_i[t] = \frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i t}) \quad (29)$$

Then, the metallicity for delay case at  $t = T$  will be

$$\begin{aligned} Z^\dagger[T] &= \frac{y(1-R)}{a} + (Z_i[T] - \frac{y(1-R)}{a})e^{-a\epsilon_0 T^\dagger} \\ &= \frac{y(1-R)}{a} + (\frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i(T-T^\dagger)}) - \frac{y(1-R)}{a})e^{-a\epsilon_0 T^\dagger} \end{aligned} \quad (30)$$

Again, we now compare the default  $Z[T]$  with  $Z^\dagger[T]$ , let's say we expect  $Z[T] > Z^\dagger[T]$ :

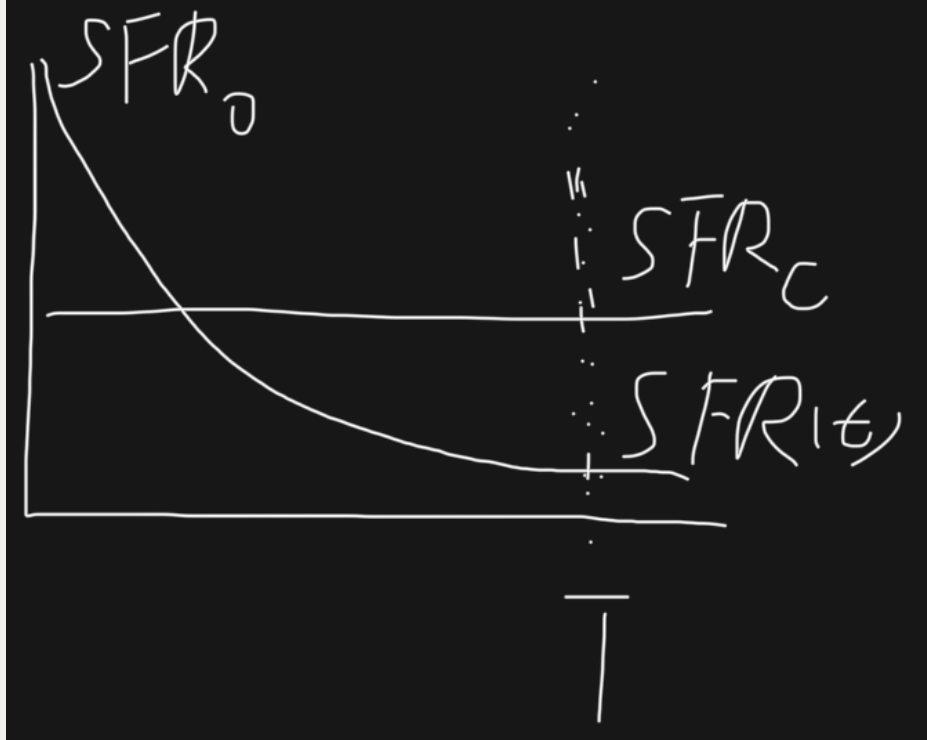
$$\begin{aligned} Z[T] &> Z^\dagger[T] \\ \Leftrightarrow \frac{y(1-R)}{a} (1 - e^{-a\epsilon_0 T}) &> \frac{y(1-R)}{a} + (\frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i(T-T^\dagger)}) - \frac{y(1-R)}{a})e^{-a\epsilon_0 T^\dagger} \\ \Leftrightarrow \frac{1}{a} - \frac{1}{a}e^{-a\epsilon_0 T} &> \frac{1}{a} + (\frac{1}{1-R+w} - \frac{1}{1-R+w}e^{-(1-R+w)\epsilon_i(T-T^\dagger)} - \frac{1}{a})e^{-a\epsilon_0 T^\dagger} \\ \Leftrightarrow -\frac{1}{a}e^{-a\epsilon_0(T-T^\dagger)} &> \frac{1}{1-R+w} - \frac{1}{1-R+w}e^{-(1-R+w)\epsilon_i(T-T^\dagger)} - \frac{1}{a} \\ \Leftrightarrow \frac{1}{a}(1 - e^{-a\epsilon_0(T-T^\dagger)}) &> \frac{1}{1-R+w}(1 - e^{-(1-R+w)\epsilon_i(T-T^\dagger)}) \\ \Leftrightarrow \frac{1}{a}(1 - e^{-a\epsilon_i(T-T^\dagger)}) &> \frac{1}{1-R+w}(1 - e^{-(1-R+w)\epsilon_i(T-T^\dagger)}) \\ \Leftrightarrow a < 1-R+w \end{aligned} \quad ($$

Which is the relation we define previously, so we indeed have  $Z[T] > Z^\dagger[T]$ .

Therefore, in case 2, we also have the anti-correlation:

$$\begin{cases} \text{SFR}[T] < \text{SFR}^\dagger[T] \\ Z(T) > Z^\dagger(T) \end{cases} \quad (32)$$

### DFD13 Case 3



In this case, we compare the default exponential decay SFH with the constant SFH.

Again, we have  $\text{SFR}[T] < \text{SFR}_c < \text{SFR}_0$ , and thus  $\epsilon_c < \epsilon_0$ .

Similarly, at  $t = T$ , we have

$$\begin{aligned}
 M_* &= \int_0^T \text{SFR}[t] dt = \int_0^T \text{SFR}_c dt \\
 &\Leftrightarrow \frac{\text{SFR}_0}{k} (1 - e^{-kT}) = \text{SFR}_c T \\
 &\Leftrightarrow \frac{\epsilon_0}{k} (1 - e^{-kT}) = \epsilon_c T
 \end{aligned} \tag{33}$$

Then, we compare the default  $Z[T]$  with  $Z_c[T]$ . let's say we expect  $Z[T] > Z_c[T]$ :

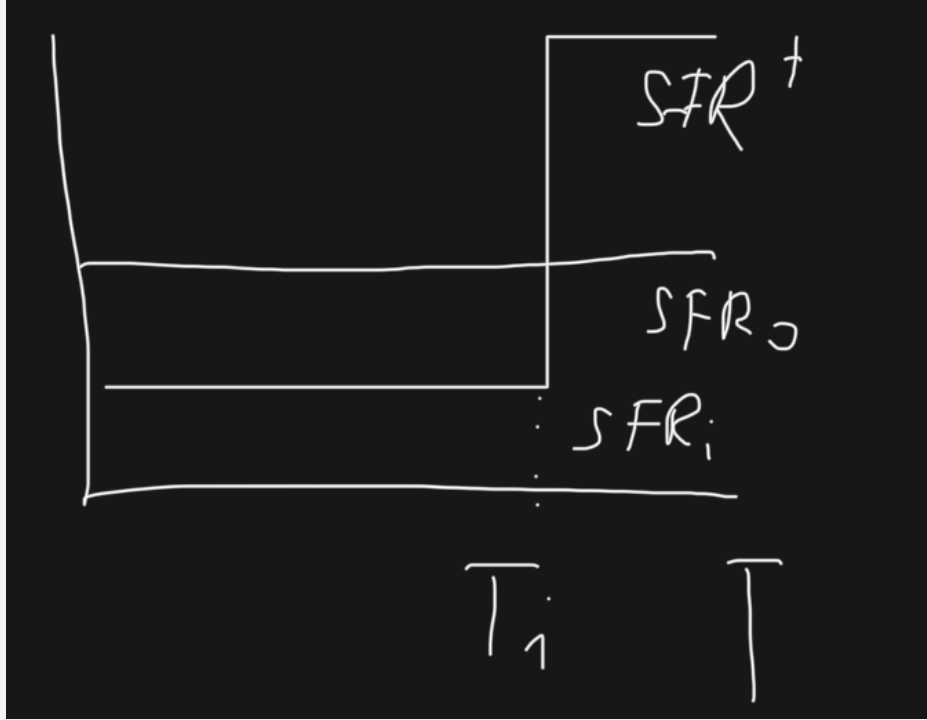
$$\begin{aligned}
 Z[T] &> Z_c[T] \\
 &\Leftrightarrow \frac{y(1-R)}{a} (1 - e^{-a\epsilon_0 T}) > \frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c T}) \\
 &\Leftrightarrow \frac{1}{a} (1 - e^{-a\epsilon_0 T}) > \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c T}) \\
 &\Leftrightarrow \frac{1}{a} (1 - e^{-a\epsilon_c T}) > \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c T}) \\
 &\Leftrightarrow a < 1 - R + w
 \end{aligned} \tag{34}$$

Which is the relation we define previously, so we indeed have  $Z[T] > Z_c[T]$ .

Therefore, in case 3, we also have the anti-correlation:

$$\begin{cases} \text{SFR}[T] < \text{SFR}_c \\ Z(T) > Z_c \end{cases} \tag{35}$$

#### DFD13 Case 4



In this case, we propose a constant SFH and and step function SFH:

$$\text{SFR}^\dagger[t] = \begin{cases} \text{SFR}_i, & t < T_i \\ \text{SFR}_f, & t \geq T_i \end{cases} \tag{36}$$

Agian, we have  $\text{SFR}_i < \text{SFR}_0 < \text{SFR}_f$  and thus  $\epsilon_i < \epsilon_0 < \epsilon_f$ .

Similarly, at  $t = T$ , we have

$$\begin{aligned}
M_* &= \text{SFR}_0 T = \text{SFR}_i T_i + \text{SFR}_f (T - T_i) \\
\Leftrightarrow (\text{SFR}_f - \text{SFR}_i) T_i &= (\text{SFR}^\dagger - \text{SFR}_0) T \\
\Leftrightarrow (\epsilon_f - \epsilon_i) T_i &= (\epsilon_f - \epsilon_0) T
\end{aligned} \tag{37}$$

Then, we compare the constant SFH's  $Z[T]$  with step function SFH's  $Z^\dagger[T]$ . let's say we expect  $Z[T] = Z_c[T]$ :

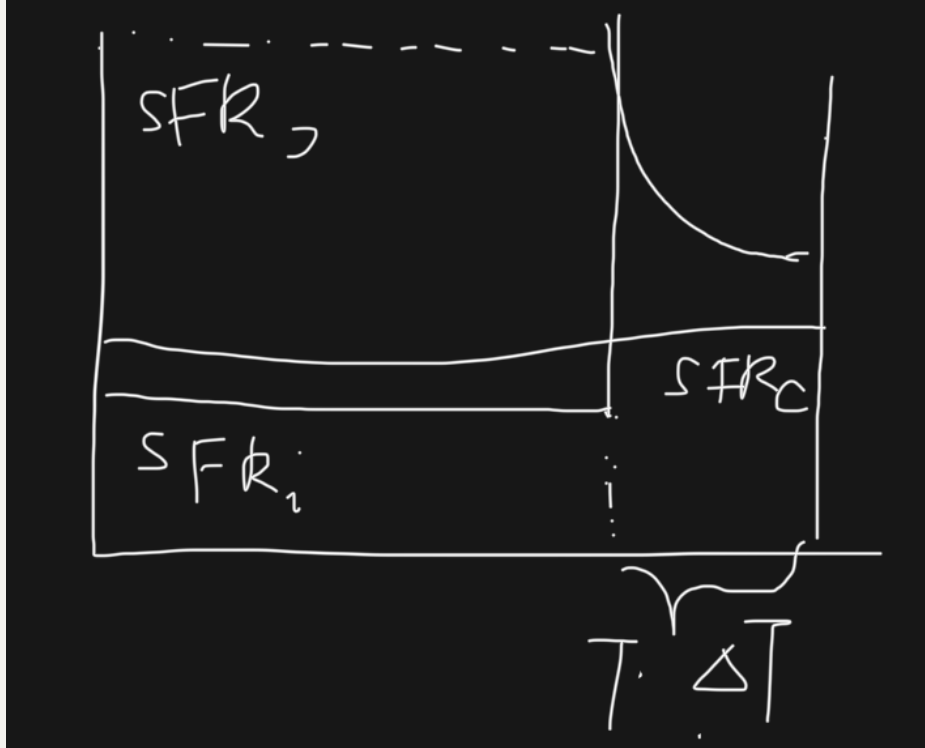
$$\begin{aligned}
Z[T] &= Z^\dagger[T] \\
\Leftrightarrow \frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_0 T}) &= \frac{y(1-R)}{1-R+w} + \left( \frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i T_i}) - \frac{y(1-R)}{1-R+w} \right) e^{-} \\
\Leftrightarrow 1 - e^{-(1-R+w)\epsilon_0 T} &= 1 + (1 - e^{-(1-R+w)\epsilon_i T_i} - 1) e^{-(1-R+w)\epsilon_f (T-T_i)} \\
\Leftrightarrow e^{-(1-R+w)\epsilon_0 T} &= e^{-(1-R+w)\epsilon_i T_i - (1-R+w)\epsilon_f (T-T_i)} \\
\Leftrightarrow -\epsilon_0 T &= -\epsilon_i T_i - \epsilon_f (T - T_i) \\
\Leftrightarrow (\epsilon_i - \epsilon_f) T_i &= (\epsilon_0 - \epsilon_f) T
\end{aligned}$$

Which is the relation we define previously, so we indeed have  $Z[T] = Z_c[T]$ .

Therefore, in case 4, we will have the no correlation:

$$\begin{cases} \text{SFR}[T] < \text{SFR}^\dagger[T] \\ Z(T) = Z^\dagger[T] \end{cases} \tag{39}$$

#### DFD13 Case 5



In this case, we have a constant SFH and a delay exponential decay SFH:

$$\text{SFR}^\dagger[t] = \begin{cases} \text{SFR}_i, & t < T \\ \text{SFR}_0 e^{-kt}, & t \geq T \end{cases} \tag{40}$$

Again, we have  $\text{SFR}_i < \text{SFR}_c < \text{SFR}_0$  and thus  $\epsilon_i < \epsilon_c < \epsilon_0$ .

Similarly, at  $t = T + \Delta T$ , we have



$$\begin{aligned}
M_* &= \text{SFR}_c(T + \Delta T) = \text{SFR}_i T + \int_0^{\Delta T} \text{SFR}_0 e^{-kt} dt \\
&= \text{SFR}_i T + \frac{\text{SFR}_0}{k} (1 - e^{-k\Delta T}) \\
&\Leftrightarrow \epsilon_c(T + \Delta T) = \epsilon_i T + (1 - R + w - a)(1 - e^{-(1-R+w-a)\epsilon_0 \Delta T})
\end{aligned} \tag{41}$$

Then, we compare the constant SFH's  $Z[T]$  with the delay SFH's  $Z^\dagger[T]$ . let's say we expect  $Z[T] > Z^\dagger[T]$ :

$$\begin{aligned}
&Z[T] > Z^\dagger[T] \\
&\Leftrightarrow \frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c(T+\Delta T)}) > \frac{y(1-R)}{a} + \left( \frac{y(1-R)}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i T}) - \frac{y(1-R)}{a} \right) \epsilon \\
&\Leftrightarrow \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c(T+\Delta T)}) > \frac{1}{a} + \left( \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i T}) - \frac{1}{a} \right) e^{-a\epsilon_0 \Delta T} \\
&\Leftarrow \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c(T+\Delta T)}) > \frac{1}{a} + \left( \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i T}) - \frac{1}{a} \right) \\
&\quad \Leftarrow \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_c T}) > \frac{1}{1-R+w} (1 - e^{-(1-R+w)\epsilon_i T}) \\
&\quad \Leftrightarrow 1 - e^{-(1-R+w)\epsilon_c T} > 1 - e^{-(1-R+w)\epsilon_i T} \\
&\quad \Leftrightarrow \epsilon_c > \epsilon_i
\end{aligned}$$

Which is the relation we define previously, so we indeed have  $Z[T] > Z^\dagger[T]$ .

Therefore, in case 5, we also have the anti-correlation:

$$\begin{cases} \text{SFR}[T] < \text{SFR}^\dagger[T] \\ Z(T) > Z^\dagger(T) \end{cases} \tag{43}$$

## WL21

In DFD13, they assume that the star formation efficiency  $\epsilon$  is constant and the inflow rate  $\Phi$  is linearly proportional to SFR. However, WL21 propose 2 cases: 1) constant  $\epsilon$  and time-variant  $\Phi$  can lead to negative correlation; 2) time-variant  $\epsilon$  and constant  $\Phi$  can lead to positive correlation.

However, I notice that their positive or negative correlations emerge based on sampling the pairs of SFR and metallicity without assuming the same  $M_*$  condition. I think I need to go through their derivation and verify this point. I personally want to take the time-variant  $\epsilon$  and constant  $\Phi$  case as the standard and ignore the other case.